

Roll No. ....

**TMA-307**

**B. TECH. (PE) (THIRD SEMESTER)  
MID SEMESTER EXAMINATION, Oct., 2022**

**ADVANCED MATHEMATICS**

**Time : 1½ Hours**

**Maximum Marks : 50**

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.  
(ii) Each sub-question carries 10 marks.

1. (a) Define analytic function. Show that the function  $f(z) = |xy|^{\frac{1}{2}}$  is not analytic at the origin. (CO2)

OR

- (b) Find the Fourier cosine transform of the function : (CO1)

$$f(x) = \begin{cases} x^2, & 0 \leq x \leq a \\ 0, & x \geq a \end{cases}$$

2. (a) Evaluate  $\int_C (12z^2 - 4iz) dz$  along the curve C joining the points (1, 1) and (2, 3). (CO3)

OR

- (b) Find all values of  $k$  such that the function  $f(z) = e^x[\cos ky + i \sin ky]$  is analytic. (CO2)

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(2)

3. (a) Prove that  $f(z) = 2x - x^3 + 3xy^2$  is harmonic function and find its harmonic conjugate. Also find the corresponding analytic function.

(CO2)

OR

- (b) Evaluate  $\oint (5z^4 - 3z^2 + 6) dz$ , where C is : (CO3)

(i) the unit circle  $|z| = 1$ .

(ii) the square ABCD with  $A = (0, 0)$ ,  $B = (1, 0)$ ,  $C = (1, 1)$  and  $D = (0, 1)$ .

(iii) the parabola  $C_1 : y = x^2$  and the parabola  $C_2 : x = y^2$  joining  $(0, 0)$  and  $(1, 1)$ .

4. (a) Determine whether the function  $f(z) = z|z|$  is analytic or not. (CO2)

OR

- (b) Using Cauchy's integral formula to compute the value of the integral

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz, \text{ where } C : |x + iy| = 3. \quad (\text{CO3})$$

5. (a) Using Cauchy's integral formula, evaluate  $\int_C \frac{2z^2 + z}{z^2 - 1}$ , where C is the circle  $|z - 1| = 1$ . (CO3)

OR

- (b) Find the function  $f(x)$  if Fourier sine transform is defined as : (CO1)

$$F_s(s) = \begin{cases} 1, & 0 \leq s \leq 1 \\ 2, & 1 \leq s \leq 2 \\ 0, & s > 2 \end{cases}$$